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Lab 1 ~ variables

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Section BSIT 2-2N

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▷ given code:

```
public class Student {
    static String school = "Pup Aozon City";
    String name;
    int grade;

    public void display () {
        double gwa = 1.75;
        System.out.println(name + " " + grade + " " + gwa + " " + school);
    }
}
```

public static void main (String[] args) {

Student s1 = new Student ();

Student s2 = new Student ();

s1.name = "Ana";

s1.grade = 90;

s2 = s1;

s2.name = "Mark";

school = "Pup - Sta. Mesa";

s1 = null;

s2.display ();

}

Part B explanation

statement

continuation

what happens

memory impact

1 s1 = new Student (); 1 creates student 1 (object in heap) 1 memory allocated on stack for s1

2 s2 = new Student (); 2 creates student 2 (object in heap) 2 memory allocated on stack for s2

3 s2 = s1 3 the content of s1 is being pointed by s2, but the objects still exist. 3 both object points to the same object.

4 s1 = null; 4 the content being pointed by s1 is terminated, but the s2 reference is still intact.

5 school = "Pup - Sta. Mesa" 5 the value of school was manipulated, since the value is now Pup - Sta. Mesa and it's in the main method, it replaced the old value. 5 the value was changed since it's on the method area.

▷ Part C - output

~~Mark 90 1.75 Pup - Sta. Mesa~~

Mark 90 1.75 Pup - Sta. Mesa

Part A = Variable

Part B = explanation (disregard)

var Name	Data type	var type	stored in	Statement	What happens	Memory impact
school	Static String	Class	Method Area	Student s1 = new Student ();	creates object 1 (student 1)	
name	String	Instance	Heap	s2 = new Student ();	creates object 2 (student 2)	
grade	int	Instance	Heap	s2 = s1	the s2 points to the content of s1 now, instead of it's content	
s1, s2	Student	local variable	Stack	s1 = null;		
gwa	double	local variable	Stack	school = "Pup - Sta. Mesa";		

Mark 90 1.75 PUP - STA.MESA

[Program finished]

